

ESG Data Dictionary

This document describes the business meaning of the ESG Analytics Assistant dataset. It is intended to be indexed by the RAG pipeline so the SQL generation model can understand business terminology, dataset semantics, analytical intent, and metric usage.

Facility_ID

Business Description: Unique identifier for each manufacturing facility.

Data Type: Integer

Classification: Identifier

Recommended Aggregation: Not aggregated

Business Notes: Primary identifier for drill-down.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Facility_Name

Business Description: Business name of the manufacturing facility.

Data Type: Text

Classification: Dimension

Recommended Aggregation: GROUP BY

Business Notes: Used in rankings and facility analysis.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Country

Business Description: Country where the facility operates.

Data Type: Text

Classification: Dimension

Recommended Aggregation: GROUP BY

Business Notes: Supports regional reporting.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

City

Business Description: City where the facility operates.

Data Type: Text

Classification: Dimension

Recommended Aggregation: GROUP BY

Business Notes: Supports geographic drill-down.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Industry

Business Description: Industry classification.

Data Type: Text

Classification: Dimension

Recommended Aggregation: GROUP BY

Business Notes: Used for industry comparisons.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Fiscal_Year

Business Description: Reporting fiscal year.

Data Type: Integer

Classification: Time

Recommended Aggregation: GROUP BY

Business Notes: Primary trend dimension.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Scope1_Emissions

Business Description: Direct greenhouse gas emissions.

Data Type: Decimal

Classification: Metric (tCO₂e)

Recommended Aggregation: SUM

Business Notes: Operational emissions.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Scope2_Emissions

Business Description: Indirect purchased electricity emissions.

Data Type: Decimal

Classification: Metric (tCO₂e)

Recommended Aggregation: SUM

Business Notes: Energy-related emissions.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Scope3_Emissions

Business Description: Value-chain emissions.

Data Type: Decimal

Classification: Metric (tCO₂e)

Recommended Aggregation: SUM

Business Notes: Largest carbon footprint component.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Energy_Consumption_MWh

Business Description: Total annual energy consumption.

Data Type: Decimal

Classification: Metric

Recommended Aggregation: SUM

Business Notes: Energy performance.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Renewable_Energy_Percentage

Business Description: Share of renewable energy.

Data Type: Decimal

Classification: Percentage

Recommended Aggregation: AVG

Business Notes: Renewable adoption KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Water_Withdrawal_ML

Business Description: Water withdrawn.

Data Type: Decimal

Classification: Metric

Recommended Aggregation: SUM

Business Notes: Water resource usage.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Water_Recycled_Percentage

Business Description: Share of water recycled.

Data Type: Decimal

Classification: Percentage

Recommended Aggregation: AVG

Business Notes: Water sustainability KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Waste_Generated_Tons

Business Description: Waste generated.

Data Type: Decimal

Classification: Metric

Recommended Aggregation: SUM

Business Notes: Waste management KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Waste_Recycled_Percentage

Business Description: Waste recycled.

Data Type: Decimal

Classification: Percentage

Recommended Aggregation: AVG

Business Notes: Circular economy KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Employee_Count

Business Description: Number of employees.

Data Type: Integer

Classification: Metric

Recommended Aggregation: SUM

Business Notes: Workforce size.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Female_Workforce_Percentage

Business Description: Female representation.

Data Type: Decimal

Classification: Percentage

Recommended Aggregation: AVG

Business Notes: Diversity KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Training_Hours

Business Description: Training delivered.

Data Type: Decimal

Classification: Hours

Recommended Aggregation: AVG or SUM

Business Notes: Learning KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Compliance_Violations

Business Description: Compliance incidents.

Data Type: Integer

Classification: Count

Recommended Aggregation: SUM

Business Notes: Governance KPI.

Typical Questions: Compare by country, trend by fiscal year, rank facilities, analyze industry performance, drill down by facility.

Business Synonyms

- Carbon footprint → Scope1_Emissions + Scope2_Emissions + Scope3_Emissions
- Headcount → Employee_Count
- Women representation → Female_Workforce_Percentage
- Green energy → Renewable_Energy_Percentage
- Carbon emissions → Scope1/Scope2/Scope3 emissions
- Plant/Site → Facility_Name